THOMAS V THOMPSON II, PH.D.

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PROFESSIONAL SUMMARY

Software & Pipeline Engineer with over two decades of expertise architecting high-performance graphics, procedural geometry, and intuitive artist toolsets for feature animation and games. Primary creator of XGen (now integrated into Autodesk Maya as an industry standard) and 3-time Academy Scientific and Technical Award nominee. Proven track record of transforming complex technical challenges into artist-friendly tools by distilling problems down to their simplifying fundamentals. Skilled at bridging offline and real-time workflows to streamline creative pipelines.

HONORS AND IMPACT

- 3-time Academy Scientific and Technical Award Nominee (Motion Picture Academy) — Recognized for the architectural creation and industry impact of XGen, iGroom, and Bonsai.
- Leonard C. Nelson Presidential Leadership Award as the most outstanding graduate of WVIT.

TECHNICAL SKILLS

- Languages: C++, Python, C, Qt / PySide / PyQt
- Engine & Pipeline Architecture: Unreal Engine 5, Katana, Maya API, XGen, Alembic, Hair Pipelines
- Rendering & Graphics: RenderMan, Redshift, Procedural Geometry, Instancing (Hair/Fur/Geometry)
- Core Competencies: Systems Architecture, Distributed Computation, Technical Leadership/Mentoring

EXPERIENCE

Lightspeed LA / Irvine, CA — Mar 2023 - Jul 2026

Senior Tools Engineer III

- Led technical integration of narrative cinematics within the game.
- Designed and created several custom tools for importing and updating sequencer content.
- Developed hair tools to convert XGen grooms to tileable flow maps for efficient fur shaders.
- Streamlined studio-wide rendering workflows for frame accurate cinematic exports.
- Implemented tools for Vcam recording, capture, and export to shot cameras.

Imagendary Studios / Irvine, CA — Feb 2022 - Jan 2023

Senior Core Software Engineer

- Technical lead of “Core Team” initiatives to design cinematic workflows, data asset pipelines, remote production workflows, and DCC integrations.
- Engineered tools to export XGen grooms and shot caches from Maya for use in Unreal Engine 5.

Blizzard Entertainment / Irvine, CA Mar 2012 - Feb 2022

Senior Software Engineer II

- Architected Belaro, an end-to-end groom-agnostic hair rendering and animation pipeline.
- Designed MobBoss, a rendering pipeline for instanced, animated, and modifiable crowds.
- Engineered the Scatter toolset, a cross-department rendering solution for mass instancing of geometry.
- Authored render-time auto-instancing for complex Katana scenes to reduce the memory footprint.
- Subject matter expert on procedural geometry and complexity management.
- Served as primary studio liaison to Autodesk, Pixar, and Redshift for XGen.

Walt Disney Animation Studios / Burbank, CA Jan 2001 - Mar 2012

Senior Development Software Engineer

- Lead architect of XGen, studio's arbitrary instancing engine used for hair, fur, grass, leaves, trees, layout, stitchwork, and much more. Now part of Autodesk Maya and used by dozens of studios.
- Patented key interpolation algorithms, "Determining Guide Connectivity", for procedural geometry.
- Delivered specialized XGen training for artists and technical directors and presented onsite at Disney Animation, Pixar, ImageMovers Digital, Digital Domain, and DisneyToon Studios.
- Designed Bonsai, a procedural forest/tree/foliage generation, animation, and rendering system.
- Authored an interactive fur grooming toolset, iGroom, to be integrated as an XGen front end.
- Built ImageToolkit as a set of unifying image manipulation libraries used by all tools.
- Software Supervisor for Lookdev/FX tools group managing 20+ people.

ADDITIONAL EXPERIENCE

- University of Utah (Research Staff): Real-time geometry computation, haptic rendering algorithms, active prototyping, distributed computation systems.
- Cray Research Inc (Intern): Vectorized rendering routines in RenderMan and optimized AVS for Cray-YMP supercomputers.
- DARPA (Intern): Designed neural networks, fuzzy logic, and hybrid algorithms in Ada to classify seismographic event data as either an earthquake or an underground nuclear test.

EDUCATION

University of Utah
Ph.D., Computer Science
Salt Lake City, UT
Dissertation: "Haptic Rendering of Trimmed NURBS Models within an Active Prototyping Environment"

Wake Forest University
M.S., Computer Science
Winston-Salem, NC
Thesis: "Three Dimensional Polygon-Mesh Morphology"

West Virginia Institute of Technology
B.S., Computer Science
Montgomery, WV
Honors: Alpha Chi Honor Fraternity, Presidential Leadership Award